

Photo capture protocols

Reference shoot guides for the Recycled Sound device register. Pick the section that matches the device shape you're capturing - you only need ONE set per device.

One set per shape class. The two BTE protocols below cover the same 6 anatomical orientations - they're not additive. Default to the **hand-held naturals** flow. The table rig is legacy: kept for reference and for the rare case where you specifically want a clean scale-card shot for register display, but real-world testing showed hand-shots outperformed calibrated rig shots for training data (22/22 naturals worked; ~3 scaled-protocol attempts failed). Non-BTE shapes and accessories ARE additional - take those when you encounter them.

1. BTE hand-held (default)

2. BTE table rig (legacy)

3. Non-BTE shapes

4. Accessories

1. BTE hand-held naturals

Hand-shot photos of BTE aids held naturally between thumb and index finger. This is the recommended protocol for training data: natural hand-shots match the deployment distribution better than calibrated rig shots.

Why this beats the table rig: in real testing, hand-shot naturals outperformed calibrated table-rig capture for training data. The scanner sees devices in volunteers' hands, not on tables - so train it that way. 22/22 naturals worked; ~3 scaled-protocol attempts failed.

Shoot order: L aid first (all 6 orientations, shots 1-6), then R aid (shots 7-12). Shoot a couple of extra angle variants of the medial (label) face if the print is hard to read - natural variation helps the model.

Anatomical naming: medial = inner (touches head, label-side); lateral = outer; anterior = front edge; posterior = back edge; superior = top end (hook); inferior = bottom end (shell base).

Lighting: indoor diffuse light. Avoid harsh shadows from overhead bulbs - face a window or use a soft lamp.

Framing: phone camera at arm's length, device fills roughly 60% of frame. Hand is the scale reference - no card needed.

1. Lateral face up

L: shot 1 . R: shot 7

Outer shell faces camera. No label visible.

HAND GRIP VIEW

2. Medial face up

L: shot 2 . R: shot 8

Inner side faces camera.
Label visible. Most important shot.

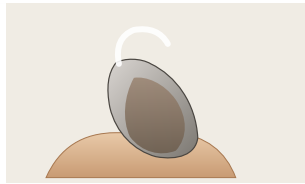
HAND GRIP VIEW

3. Anterior edge up

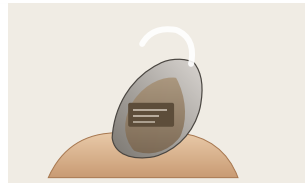
L: shot 3 . R: shot 9

Front edge of device faces camera. Hook curves away.

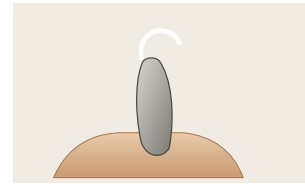
HAND GRIP VIEW



Pinch shell at posterior edge; lateral face points up at camera.



Flip so label faces up. This shot drives OCR brand/model detection - get it sharp.



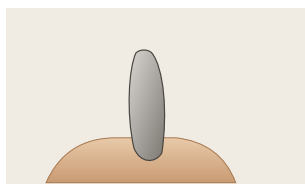
Hold by superior/inferior ends so front edge is exposed.

4. Posterior edge up

L: shot 4 . R: shot 10

Back edge faces camera. Hook hidden behind body.

HAND GRIP VIEW



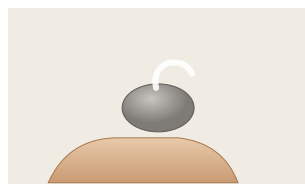
Same grip as #3 but rotated 180 degrees. No hook silhouette.

5. Superior end up

L: shot 5 . R: shot 11

Hook end points at camera. Top-down silhouette of body.

HAND GRIP VIEW



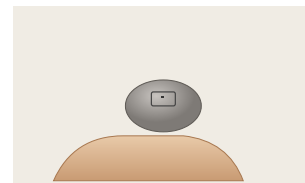
Hold by inferior end; aim camera down the long axis from the hook side.

6. Inferior end up

L: shot 6 . R: shot 12

Battery-door end points at camera. Bottom cross-section.

HAND GRIP VIEW



Hold by superior end (or hook); shoot down the long axis. Battery door visible.

2. BTE table rig (legacy - skip unless you need a scale-card reference)

6 orientations x 2 aids (L then R) = 12 shots per pair. Device on the table, credit card in frame as scale reference. Camera always straight down.

Legacy protocol. Use the [hand-held flow](#) instead - it covers the same 6 anatomical orientations and beat this rig in real testing. Only fire this protocol when you specifically need a calibrated scale-card shot (e.g. for register display where the audiologist wants a clean overhead photo). Do NOT shoot this AND the hand-held set on the same device - they're redundant.

Shoot order: Always L aid first (all 6 orientations 1-6), then R aid (all 6 orientations 7-12). Same anatomical axis sequence both times.

Anatomical naming: medial = inner (touches head, label-side); lateral = outer; anterior = front edge; posterior = back edge; superior = top end (hook); inferior = bottom end (shell base).

1. Lateral face up

L: shot 1 . R: shot 7

Outer shell up. No label visible.

CAMERA VIEW

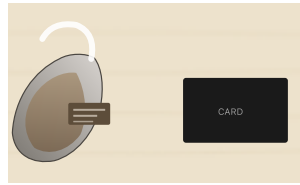


2. Medial face up

L: shot 2 . R: shot 8

Inner side up. **Label visible.**

CAMERA VIEW

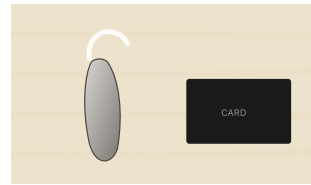


3. Anterior edge up

L: shot 3 . R: shot 9

Device on its posterior edge. Front edge faces camera.

CAMERA VIEW



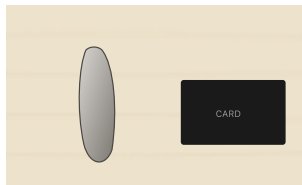
Thin edge-on silhouette - narrow profile, hook visible curving away.

4. Posterior edge up

L: shot 4 . R: shot 10

Device on its anterior edge. Back edge faces camera.

CAMERA VIEW



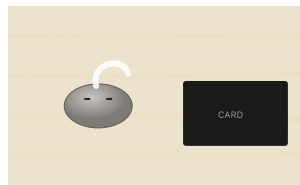
Same thin silhouette, hook hidden behind (no protrusion).

5. Superior end up

L: shot 5 . R: shot 11

Device standing on inferior end. Hook attachment faces camera.

CAMERA VIEW



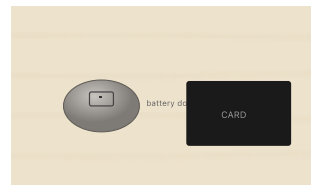
Oval cross-section (wider than tall) with hook stub.

6. Inferior end up

L: shot 6 . R: shot 12

Device standing on superior end. Bottom of shell faces camera.

CAMERA VIEW



Full cross-section showing the battery door.

Card visible in every shot. Some orientations are physically awkward - BTEs don't balance well on edges or ends - just hold them with a finger and shoot whatever silhouette you can.

3. Non-BTE shapes - RIC, ITE, CIC, IIC

Smaller in-ear devices. Different anatomy, different shot lists. Hand-held naturals for all; plain background where possible.

General rules: hand-held, indoor diffuse light, fill ~60% of frame. Use macro mode if your phone supports it for CIC/IIC. Coin for scale on IIC/CIC if you want a reference (optional).

RIC (Receiver-in-Canal)

9 shots per aid

ITE (In-the-Ear)

4 shots per aid

CIC (Completely-in-Canal)

3 shots per aid

Looks like a BTE but with a thin wire ending in a receiver-dome that sits in the ear canal.

Body (BTE-style): all 6 BTE orientations from the hand-held protocol above (lateral, medial, anterior, posterior, superior, inferior).

Receiver bud:

- Shot 7 - receiver front (looking at the dome end)
- Shot 8 - receiver side profile

Wire:

- Shot 9 - full wire path from body to receiver, laid flat

Pinch the body to expose the wire-exit point in shot 9; keep wire from kinking.

Custom-moulded faceplate, fills the concha. No external wire.

- Shot 1 - faceplate up (controls, vent, mic ports visible)
- Shot 2 - canal side (the side that goes into the ear, receiver port visible)
- Shot 3 - lateral profile (side-on, faceplate edge-on left)
- Shot 4 - posterior profile (rotated 180 degrees from #3)

Hold by the canal end so the faceplate is unobstructed in shot 1. The faceplate is where the brand/serial usually prints.

Tiny, sits fully in the canal. Removal handle (clear filament) usually visible.

- Shot 1 - faceplate end (looking at the outer end, removal handle visible)
- Shot 2 - side profile (long axis horizontal)
- Shot 3 - close-up of removal handle / pull-cord

Use macro mode if available. A coin in frame adds scale reference (optional).

IIC (Invisible-in-Canal)

2 shots per aid

Deepest-fitting style. Smaller than CIC. Often no visible controls.

- Shot 1 - faceplate end (outer end, may show removal filament)
- Shot 2 - canal end (inner end, receiver port)

Macro mode strongly recommended. Plain matte background (paper, cloth) so the device silhouette is clean.

4. Accessories

Non-aid items in the register: chargers, charging cases, domes, wax filters, tubing, remote controls. Plain background, hand-held or flat.

General rule per accessory: 2 shots - top view + side profile, on a plain background. Special cases below add shots.

End of session: one group shot of any matched set (e.g. charger + aid + charging cable together).

Chargers (standalone)

2 shots

Wall chargers, USB bricks, charging pads.

- Shot 1 - top view (logos, model markings)
- Shot 2 - side profile (port, cable exit)

Charging cases

4 shots

Combined charger + storage case for the aids.

- Shot 1 - closed, top view
- Shot 2 - closed, side profile
- Shot 3 - open, empty (charging contacts visible)
- Shot 4 - open, with aids seated

Keep case on a flat plain surface; camera straight down for shots 1 and 3/4.

Domes

2 shots + scale

Silicone tips for RIC/BTE receivers (open, closed, power, double).

- Shot 1 - top view (opening visible if vented)
- Shot 2 - side profile
- Include a coin in frame (10c or 20c piece) for scale - domes are tiny.

Wax filters

2 shots + scale

Tiny replaceable filters that sit at the receiver outlet.

- Shot 1 - top view of the carrier/wheel (filters in slots)
- Shot 2 - single filter close-up
- Coin in frame for scale.

Macro mode strongly recommended.

Tubing

2 shots

Slim tubes or standard tubes that connect BTE hook to ear mould.

- Shot 1 - laid flat, full length (note coil if pre-curved)
- Shot 2 - close-up of connector end

Remote controls

2 shots

Hand-held remotes that pair with aids.

- Shot 1 - top view (buttons, display, brand)
- Shot 2 - back / battery compartment

Matched set group shot

1 shot

Final shot of the session for any matched accessory + aid set.

- Arrange charger, cable, aids, and any included accessories on a plain background.
- Top-down camera view; everything in one frame.

This shot documents what came as a kit - useful for the donor record and the audiologist's QA check.

Naming and storage: raw photos go to `data/photos_nick_aids/<DEVICE_ID>/`. Anatomical orientation is encoded in the `manifest.json` sidecar, not the filename - that way we can re-tag without renaming files.